

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M.Charan(192525167) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

d) z-score normalization

$$z = \frac{x - \mu}{\sigma}$$

Age	z
23	$\frac{23 - 46.44}{12.85} = \frac{-23.44}{12.85} = -1.82$
23	$23 - 46.44 / 12.85 = -1.82$
27	$27 - 46.44 / 12.85 = -1.51$
27	$27 - 46.44 / 12.85 = -1.51$
39	$39 - 46.44 / 12.85 = -0.58$
41	$41 - 46.44 / 12.85 = \frac{-5.44}{12.85} = -0.42$
47	$47 - 46.44 / 12.85 = \frac{0.56}{12.85} = 0.04$
49	$\frac{49 - 46.44}{12.85} = \frac{2.56}{12.85} = 0.20$
50	$\frac{50 - 46.44}{12.85} = \frac{3.56}{12.85} = 0.28$
52	$\frac{52 - 46.44}{12.85} = \frac{5.56}{12.85} = 0.43$

$$\frac{54 - 46.44}{12.85} = \frac{7.56}{12.85} = 0.59$$

$$\frac{56 - 46.44}{12.85} = \frac{9.56}{12.85} = 0.74$$

$$\frac{57 - 46.44}{12.85} = \frac{10.56}{12.85} = 0.82$$

$$\frac{58 - 46.44}{12.85} = \frac{11.56}{12.85} = 0.90$$

$$\frac{60 - 46.44}{12.85} = \frac{13.56}{12.85} = 1.06$$

$$\frac{61 - 46.44}{12.85} = \frac{14.56}{12.85} = 1.13$$

E) formula

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

$$\sum x = 836$$

$$\sum y = 518.1$$

$$\sum x^2 = 41898$$

$$\sum y^2 = 16368.59$$

Suppose a group of 12 sales price records have been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 72, 92, 204, 215. Partition them into three bins by each of the following methods: (a) equal-width partitioning (b) clustering.

(a) equal-frequency partitioning.

* total no. of records = 12

* total no. of bins = 3

equal-frequency partitioning

$$\text{records per bin} = \frac{12}{3} \left[\frac{\text{total}}{\text{no. of bins}} \right] = 4$$

∴ each bin contains 4 values

Bin 1

5, 10, 11, 13

Bin 2

15, 35, 50, 55

Bin 3

72, 92, 204, 215

b) equal partitioning

$$\frac{\text{max} - \text{min}}{\text{bins}} =$$

The intervals are

Bin 1: 5-74

Bin 2: 75-144

Bin 3: 145-215

Bin 1

5-74

Bin 2

75-144

Bin 3

145-215

c) clustering

cluster 1

5, 10, 11, 13, 15

cluster 2

35, 50, 55, 72, 92

cluster 3

204, 215

* Suppose a group of hospital tested the age and body fat data for 18 randomly selected adults with the following result.

a) calculate the mean, median and standard deviation of age and % fat.

b) Draw the box plots for age and % fat.

a) a scatter

mean total no. of observations

$$\frac{23+23+27+27+39+41+43}{18}$$

$$= \frac{236}{18} = 13.11$$

median 9th = 50

10th = 52

$$= \frac{50+52}{2} = 51$$

mean (% fat) = $\frac{9.5 + 26.5 + 7.8 + 17.8 + 31.4 + 25.9 + 27.4 + 27.2 + 31.2}{18}$

$$= \frac{518.1}{18} = 28.78$$

$$SD = 12.85$$

median (% fat)

9th value = 30.2

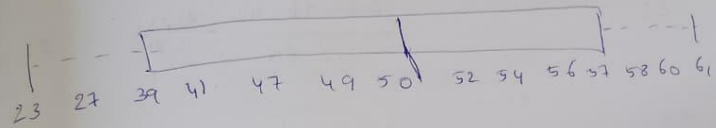
10th value = 31.2

$$\frac{30.2 + 31.2}{2} = \frac{61.4}{2} = 30.7$$

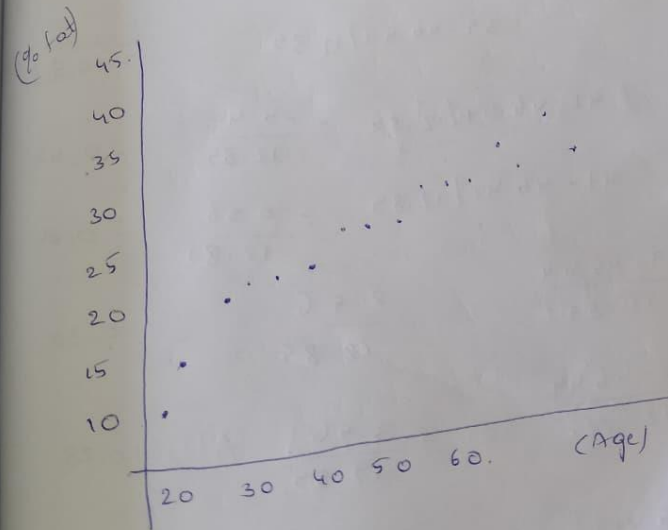
b) min = 23

$$\frac{52+54+54+56+57+58+58+60+61}{9}$$

$$Q3 = 57$$



c) Scatter plot -



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
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